Strand 2: Algebra

Sub-strand 3: Variables and Equations

Lesson 1: Linear Equations, Inequalities, and Number-Line Solutions

1. Curriculum Standard and Overview

- Strand: 2 — Algebra
- Sub-strand: 3 — Variables and Equations
- Content Standard: B9.2.3.1
- Indicators:
- B9.2.3.1.1 — Solve single-variable linear equations and inequalities with rational coefficients.
- B9.2.3.1.2 — Illustrate solution sets of linear inequalities on the number line.
- B9.2.3.1.3 — Solve real-life problems involving linear equations and inequalities.
- Mock-paper links: P1 Q2, Q4, Q8, Q18, Q25, Q39; P2 Q3(c), Q6(b).

2. Preamble — Equation vs Inequality

- An equation says two things are equal: 2x + 1 = 9.
- An inequality says one is bigger/smaller: 2x + 1 > 9.

Solving means: find every value of x that makes the statement true.

3. Solving Linear Equations

3.1 Golden rule

Whatever you do to one side, do to the other.

Mock P1 Q25: 13x - 12 = 5x + 60

1. Subtract 5x from both sides: 8x - 12 = 60.
2. Add 12: 8x = 72.
3. Divide by 8: x = 9.

Answer: C — 9.

3.2 Simple two-step (Mock P1 Q39)

15 - 2x = 6

1. Subtract 15: −2x = −9.
2. Divide by −2: x = 4.5.

Answer: C — 4.5.

3.3 Equations with fractions (Mock P2 Q6b)

4/(3x) = 7 - 3/x

1. Multiply every term by 3x (the LCM of the denominators):

$$4 = 21x - 9$$

2. 4 + 9 = 21x → 13 = 21x → x = 13/21.

Answer: x = 13/21.

4. Solving Inequalities

4.1 Same as equations — with one extra rule

If you multiply or divide by a negative number, flip the inequality sign.

4.2 Worked: Mock P1 Q2

2x + 1 > 9

1. Subtract 1: 2x > 8.
2. Divide by 2: x > 4.

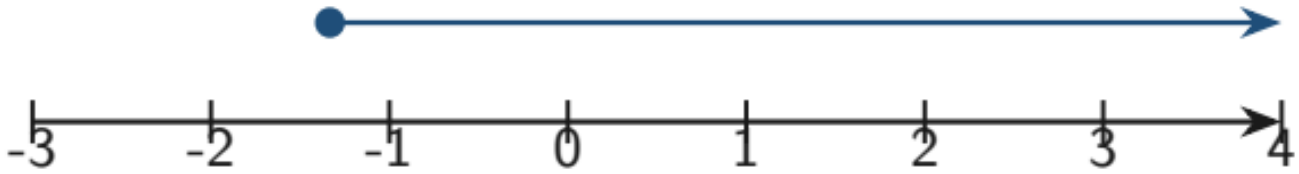
Answer: B — x > 4.

4.3 Worked: Mock P2 Q3(c)

6 ≥ 2 - 3x

1. Subtract 2: 4 ≥ −3x.
2. Divide by −3 and flip: −4/3 ≤ x, i.e. x ≥ −4/3 (or x ≥ −1 ⅓).

See Figure 1.



4.4 Brackets first (Mock P1 Q18)

5 - 5(3 - y) ≤ 65, y is one of the first 6 multiples of 5.

1. Expand: 5 - 15 + 5y ≤ 65 → −10 + 5y ≤ 65.
2. 5y ≤ 75 → y ≤ 15.
3. First 6 multiples of 5: 5, 10, 15, 20, 25, 30.
4. Those with y ≤ 15: {5, 10, 15}.

Answer: A.

5. Word Problems → Equations

5.1 Consecutive odd integers (Mock P1 Q4)

Three consecutive odd integers sum to 33.

Let them be n, n+2, n+4 (odd numbers jump by 2).

$$n + (n+2) + (n+4) = 33 \Rightarrow 3n + 6 = 33 \Rightarrow 3n = 27 \Rightarrow n = 9$$

Numbers: 9, 11, 13. Answer: A.

5.2 Age problems (Mock P1 Q8)

Kojo's age in the next 10 years will be 4 times his age 5 years ago.

Let Kojo be x now.

$$x + 10 = 4(x - 5)$$
$$x + 10 = 4x - 20$$
$$10 + 20 = 4x - x$$

$$30 = 3x \Rightarrow x = 10$$

Check: in 10 years he is 20; 5 years ago he was 5; 4 × 5 = 20 (correct).

Answer: C — 10.

6. Number-Line Representation

Symbol	Dot	Meaning
> or <	open ○	endpoint not included
≥ or ≤	solid ●	endpoint included

Arrow points toward larger numbers (to the right).

7. Twenty Fully Worked Examples

Example 1 — Mock P1 Q25

13x - 12 = 5x + 60 → x = 9. C.

Example 2 — Mock P1 Q39

15 - 2x = 6 → x = 4.5. C.

Example 3 — Mock P1 Q2

2x + 1 > 9 → x > 4. B.

Example 4 — Mock P2 Q3(c)

6 ≥ 2 - 3x → x ≥ −4/3.

Example 5 — Mock P1 Q18

5 - 5(3-y) ≤ 65 → y ≤ 15; from {5,10,15,20,25,30} → {5, 10, 15}. A.

Example 6 — Mock P2 Q6(b)

4/(3x) = 7 - 3/x → x = 13/21.

Example 7 — Mock P1 Q4

Consecutive odd integers summing to 33: 9, 11, 13. A.

Example 8 — Mock P1 Q8 (age)

x + 10 = 4(x - 5) → x = 10.

Example 9 — One-step

x + 7 = 12 → x = 5.

Example 10 — Two-step

3x - 4 = 11 → 3x = 15 → x = 5.

Example 11 — Unknowns on both sides

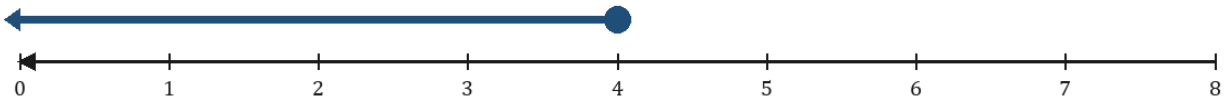
$5x + 2 = 2x + 14 \rightarrow 3x = 12 \rightarrow x = 4$.

Example 12 — Inequality with flip

$-2x > 6 \rightarrow x < -3$ (divided by -2, sign flipped).

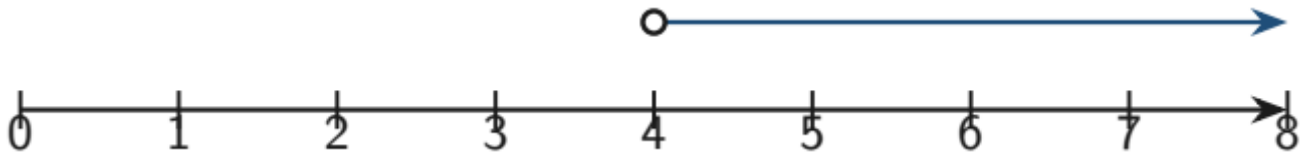
Example 13 — Inequality on number line

$x \leq 4$.



Example 13b — Strict inequality on number line

$x > 4$.



Example 14 — Strict inequality

$x < 4$.

Example 15 — Consecutive integers (not odd)

Three consecutive integers sum to 27: $n + (n+1) + (n+2) = 27 \rightarrow n = 8 \rightarrow 8, 9, 10$.

Example 16 — Rectangle perimeter equation

Perimeter 30, length is 2 more than width: $2(w + w + 2) = 30 \rightarrow w = 6.5$.

Example 17 — Fraction equation

$x/3 + 2 = 6 \rightarrow x/3 = 4 \rightarrow x = 12$.

Example 18 — Bracket equation

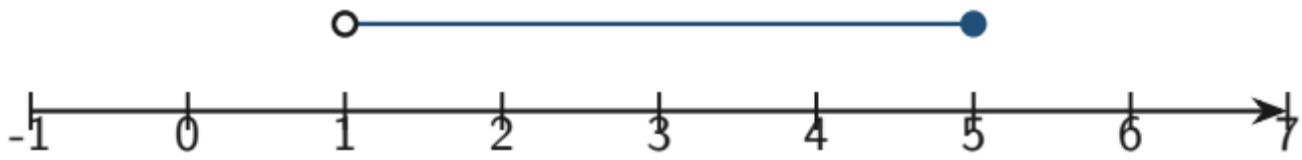
$2(x - 3) = 8 \rightarrow 2x - 6 = 8 \rightarrow x = 7$.

Example 19 — Verify a solution

Is $x = 9$ a solution of $13x - 12 = 5x + 60$? LHS = 117 - 12 = 105; RHS = 45 + 60 = 105 (correct).

Example 20 — Compound inequality

$1 < x \leq 5$.



8. Comprehensive Exercise Sets (40 Questions)

Work through these in order. Questions marked [Mock 2026] come from the BECE mock paper.

Exercise 1 — Basic equations (includes mock)

1. [Mock 2026 P1 Q39] Solve 15 - 2x = 6.
   A. −10.5
   B. −4.5
   C. 4.5
   D. 10.5
2. [Mock 2026 P1 Q25] If 13x - 12 = 5x + 60, find x.
   A. −9
   B. −6
   C. 9
   D. 6
3. Solve x + 7 = 12.
4. Solve 3x - 4 = 11.
5. Spiral review: Find 12x + 9q when x = 11, q = 3.

Exercise 2 — Unknowns on both sides

1. Solve 5x + 2 = 2x + 14.
2. Solve 7x - 5 = 3x + 15.
3. Solve 4(x - 3) = 2x.
4. Spiral review: 20xy - 5xy = ?
5. Spiral review: Evaluate 5(2x - 3) when x = 4.

Exercise 3 — Inequalities (includes mock)

1. [Mock 2026 P1 Q2] Solve the inequality 2x + 1 > 9.
   A. x > 3
   B. x > 4
   C. x > 2
   D. x > 1
2. Solve −2x > 6.
3. Solve 3x - 1 ≤ 8.
4. Solve −x < 5.
5. Spiral review: Write 5.6 × 10^(−4) as an ordinary number.

Exercise 4 — Number line (includes mock)

1. [Mock 2026 P2 Q3(c)] Find the truth set of 6 ≥ 2 - 3x and represent your result on the number line.
2. Represent x ≤ 4 on a number line.
3. Represent x < 4 on a number line. How does it differ from Question 2?
4. Represent 1 < x ≤ 5 on a number line.
5. Spiral review: Share GH₵ 720 in the ratio 2 : 3.

Exercise 5 — Bracket inequalities (includes mock)

1. [Mock 2026 P1 Q18] Find the solution set of 5 - 5(3 - y) ≤ 65, if y is one of the first 6 multiples of 5.
   A. {5, 10, 15}
   B. {5, 10}
   C. {5, 10, 15, 20}
   D. {5, 10, 15, 20, 25, 30}
2. Solve 2(x - 3) = 8.

3. Solve 3(2x + 1) = 21.
4. Spiral review: (−3)^(x-2) = −243. Find x.
5. Spiral review: Speed of a van covering 154 km in 1 ¾ h.

Exercise 6 — Fraction equations (includes mock)

1. [Mock 2026 P2 Q6(b)] Solve 4/(3x) = 7 - 3/x.
2. Solve x/3 + 2 = 6.
3. Solve 2x + 1/2 = 5.
4. Spiral review: (0.07 × 0.02) ÷ 14.
5. Spiral review: Classify −3 under ℕ, 𝕎, ℤ, ℚ.

Exercise 7 — Word problems (includes mock)

1. [Mock 2026 P1 Q4] The sum of three consecutive odd integers is 33. Find the numbers.
   A. 9, 11 and 13
   B. 7, 9 and 11
   C. 11, 13 and 15
   D. 13, 15 and 17
2. [Mock 2026 P1 Q8] Kojo's age in the next 10 years will be 4 times his age 5 years ago. How old is Kojo now?
   A. 5
   B. 6
   C. 10
   D. 15
3. Three consecutive integers sum to 27. Find them.
4. A rectangle has perimeter 30 cm. The length is 2 cm more than the width. Find the width.
5. Spiral review: In a market of 36 fishmongers, 28 sell mackerel, 18 tilapia, 13 both. How many sell neither?

Exercise 8 — Mixed challenge

1. Verify that x = 9 satisfies 13x - 12 = 5x + 60.
2. Solve |x| = 3 (two answers).
3. Solve 2x > −4 and write the truth set on a number line.
4. Spiral review: Simplify frac{25x^(−4)}{5^(−3) × 10x × t^(−8)}.
5. Spiral review: Find the gradient of the line through (1, 2) and (4, y) if the gradient is −2.

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. −2x = −9 → x = 4.5 → C.
2. 8x = 72 → x = 9 → C.
3. x = 5.
4. 3x = 15 → x = 5.
5. 159.

Solutions to Exercise 2

1. 3x = 12 → x = 4.
2. 4x = 20 → x = 5.
3. 4x - 12 = 2x → x = 6.
4. 15xy.
5. 25.

Solutions to Exercise 3

1. $2x > 8 \rightarrow x > 4 \rightarrow$ B.
2. $x < -3$ (sign flipped).
3. $x \leq 3$.
4. $x > -5$ (sign flipped).
5. 0.00056.

Solutions to Exercise 4

1. $x \geq -4/3$ (Figure 1).
2. $x \leq 4$.
3. $x < 4$.
4. Open at 1, solid at 5, segment between.
5. GH₵ 288 and GH₵ 432.

Solutions to Exercise 5

1. $y \leq 15 \rightarrow \{5, 10, 15\} \rightarrow$ A.
2. $x = 7$.
3. $2x + 1 = 7 \rightarrow x = 3$.
4. $x = 7$.
5. 88 km/h.

Solutions to Exercise 6

1. 4 = 21x - 9 → x = mathbf{13/21}.
2. $x = 12$.
3. $2x + 1 = 10 \rightarrow x = 4.5$.
4. 0.0001.
5. $\mathbb{Z}$, $\mathbb{Q}$.

Solutions to Exercise 7

1. $3n + 6 = 33 \rightarrow n = 9 \rightarrow$ 9, 11, 13 → A.
2. $x + 10 = 4(x-5) \rightarrow x = 10 \rightarrow$ C.
3. $3n + 3 = 27 \rightarrow n = 8 \rightarrow$ 8, 9, 10.
4. w = 6.5 cm.
5. 3.

Solutions to Exercise 8

1. LHS = 105, RHS = 105 (correct).
2. $x = 3$ or $x = -3$.
3. $x > -2$.
4. mathbf{625 t^(8)/(2 x^(5))}.
5. $y = -4$.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q2	$2x + 1 > 9$	B — $x > 4$
P1 Q4	Consecutive odd integers sum 33	A — 9, 11, 13
P1 Q8	Kojo age	C — 10

P1 Q18	5 - 5(3-y) ≤ 65, multiples of 5	A — {5, 10, 15}
P1 Q25	13x - 12 = 5x + 60	C — 9
P1 Q39	15 - 2x = 6	C — 4.5
P2 Q3(c)	6 ≥ 2 - 3x	x ≥ −4/3
P2 Q6(b)	4/(3x) = 7 - 3/x	x = 13/21

11. Lesson Summary

Skill	Rule
Solve equation	Inverse operations, balance both sides
Solve inequality	Same — flip sign if ÷/× by negative
Number line	○ for <,>; ● for ≤,≥
Word problem	Let x = ldots, write equation, solve, check

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings required.